

House Price Prediction — Project Summary

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Problem

Real estate buyers and sellers often rely on guesswork to estimate fair property value. This project builds a regression model that predicts house prices from property features and identifies which features matter most.

Data & Method

545 houses, 13 features (area, bedrooms, bathrooms, stories, mainroad, guestroom, basement, hot water heating, air conditioning, parking, preferred area, furnishing status). No missing values or duplicates. Data was cleaned (yes/no → 1/0, one-hot encoding for furnishing status), then split 80/20 into train and test sets. Two models were trained: Linear Regression and Random Forest Regressor.

Results

- **Linear Regression:** MAE = 970,043 | RMSE = 1,324,507 | $R^2 = 0.6529$
- **Random Forest:** MAE = 1,028,648 | RMSE = 1,401,052 | $R^2 = 0.6116$

Key Insights

- **Top price drivers:** area, bathrooms, and air conditioning showed the strongest positive correlation with price (0.54, 0.52, 0.45), matching real-world expectations.
- **Accuracy:** Linear Regression explains ~65% of price variation, with an average error of ~970,000 per house — a useful first estimate, not a final valuation.
- **Surprise:** Random Forest slightly underperformed Linear Regression. With only 545 rows, simpler models generalize better; complex models can overfit small datasets.

Recommendation for a Real Estate Business

Use this model as a fair-pricing baseline when listing new properties. An agent can input a new house's area, bathrooms, AC, and other features to get a starting price estimate, then adjust for local market conditions — helping prevent listings from being significantly overpriced (sitting unsold) or underpriced (losing the seller money).